# The Ions in Acids & Alkalis

**Exam code:** 0620 & 0971

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## Neutralisation reactions

### What is a neutralisation reaction?

- Acids are a source of hydrogen ions, $H^+$
- Bases (or alkalis) are a sources of hydroxide ions, $OH^-$
- When they react together in a neutralisation reaction, the $H^+$ ions react with the $OH^-$ ions to produce **water**
- This is the **net ionic equation** of all acid-base neutralisations and is what leads to a neutral solution, since water has a pH of 7:

$$H^+ (aq) + OH^- (aq) \rightarrow H_2O (l)$$

- Not all reactions of acids are neutralisations:
  - For example when a metal reacts with an acid, although a salt is produced there is no water formed so it does not fit the definition of neutralisation

**Examiner Tips and Tricks**

Not all reactions of acids are neutralisations. For example, when a metal reacts with an acid, although a salt is produced there is no water formed so it does not fit the definition of neutralisation.

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

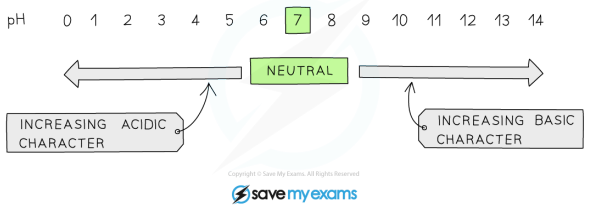
Did this video help you? Yes No

# Hydrogen ion concentration & pH

## What is the pH scale?

- The pH scale goes from 1 – 14
- All acids have pH values of **below** 7, all alkalis have pH values of **above** 7
- The **lower** the pH then the **more acidic** the solution is:
  - pH 0–2 = strong acid
    - Extremely acidic substances can have values of below 1
  - pH 3–6 = weak acid
- The **higher** the pH then the **more alkaline** the solution is:
  - pH 8–11 = weak alkali
  - pH 12–14 = strong alkali
- A solution of pH 7 is described as being **neutral**

## The pH scale

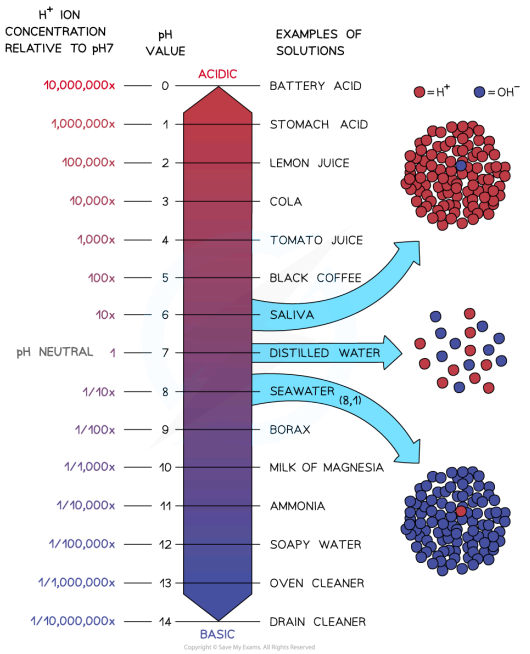



***The pH scale showing acidity, neutrality and alkalinity***

## The pH scale and hydrogen ions

- We have already seen that acids are substances that contain hydrogen ions in solution
- The more hydrogen ions the stronger the acid, but the lower the pH
- The higher the concentration of hydroxide ions in a solution the higher the pH
- So pH is a measure of the concentration of $H^+$ ions in solution, but they have an inverse relationship

## Concentration of hydrogen ions and pH




***An acid with a pH of 3 has ten times the concentration of $H^+$ ions than an acid of pH 4. An acid with a pH of 2 has 10 x 10 = 100 times the concentration of $H^+$ ions than an acid with a pH of 4***

### Examiner Tips and Tricks

Acid strength is reflected in how many hydrogen ions are in solution. The more hydrogen ions the lower the pH and vice-versa.

## How is universal indicator used?

- Universal indicator is a mixture of different **indicators** which is used to measure the pH
- A drop is added to the solution and the colour is matched with a colour chart which indicates the pH which matches specific colours

*The pH scale with the Universal Indicator colours which can be used to determine the pH of a solution*

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